S2 Supplementary Data

Complete onset/peak-phase permutation tables, per-subject Rayleigh summaries, robustness-check detail, and the full alignment-moderation family

Article type: Supplement

Short title: S2 Data - RVC_SLEEP_peak_analysis

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S2.0 Complete primary family (onset-phase, with bootstrap CI)

Sample	Channel	N	R_1	perm p (R_1)	boot R_1 95% CI	T_2	perm p (T_2)
Full catalog	phi_cardiac_onset	23,029	0.0093	0.353	[0.0022, 0.0241]	0.00007	0.082
Full catalog	phi_resp_onset	23,029	0.0189	0.151	[0.0062, 0.0404]	0.00019	0.027
Quality-filtered	phi_cardiac_onset	20,877	0.0043	0.827	[0.0018, 0.0204]	0.00002	0.116
Quality-filtered	phi_resp_onset	20,877	0.0153	0.362	[0.0037, 0.0398]	0.00012	0.075

Table 1: Complete primary family: corrected subject-rotation permutation test, spindle-onset phase, both harmonic statistics. BH-FDR q (both statistics, both channels×samples) in the main text Table 2. Source: `results/rvc_sleep_20260710/11_richsleep_maalb_corrected/molB_primary_corrected_permutation.csv`.

S2.1 Complete secondary/exploratory family (peak-phase + 190-event subsample)

Sample	Channel	N	R_1	perm p (R_1)	boot R_1 95% CI	T_2
Full catalog	phi_cardiac_peak	23,029	0.0159	0.0200	[0.0074, 0.0278]	0.00066
Full catalog	phi_resp_peak	23,029	0.0453	0.0005	[0.0236, 0.0716]	0.00217
Quality-filtered	phi_cardiac_peak	20,877	0.0145	0.0675	[0.0054, 0.0270]	0.00060
Quality-filtered	phi_resp_peak	20,877	0.0418	0.0030	[0.0182, 0.0705]	0.00189
Geometry-valid 190	phi_cardiac_onset	190	0.0504	0.692	[0.0137, 0.1872]	0.01243
Geometry-valid 190	phi_resp_onset	190	0.0210	0.929	[0.0103, 0.1528]	0.00412
Geometry-valid 190	phi_cardiac_peak	190	0.0876	0.127	[0.0278, 0.1786]	0.01681
Geometry-valid 190	phi_resp_peak	190	0.0428	0.808	[0.0129, 0.1855]	0.00915

Table 2: Complete secondary/exploratory family. The last four rows (geometry-valid 190-event subsample) reproduce the companion article's own R6b numbers on that subsample, confirming this reanalysis's headline result (main text R3) is specific to the 120x-larger full/quality-filtered catalog, not an artifact of a different statistic applied to the same small sample. Source: `results/rvc_sleep_20260710/11_richsleep_maalb_corrected/molB_secondary_exploratory_permutation.csv`.

S2.2 Per-subject Rayleigh test summary (full catalog, all 4 channels)

Channel	Median R	Median N events/subj.	Range N events/subj.	N FDR-sig./24
`phi_cardiac_onset`	0.046	1,030	161-2,562	0/24
`phi_resp_onset`	0.061	1,030	161-2,562	7/24
`phi_cardiac_peak`	0.036	1,030	161-2,562	1/24
`phi_resp_peak`	0.071	1,030	161-2,562	8/24

Table 3: Per-subject Rayleigh test (Zar 1999 approximation), summarized across the 24 subjects; full 96-row per-subject table (subject ID, N, mean angle, R, p, FDR q) in `molB_per_subject_rayleigh.csv`. Peak-phase channels show more within-subject FDR survivors than the corresponding onset-phase channels for respiration (8 vs. 7) and cardiac (1 vs. 0), consistent with a genuine, if individually modest, population-level shift rather than a purely second-order artifact. Source: `results/rvc_sleep_20260710/11_richsleep_maalb_corrected/molB_per_subject_rayleigh.csv`.

S2.3 Robustness-check detail (companion to main-text R4)

The duration-tercile (Table 3 in main text) and split-half (Table 4 in main text) tables are reproduced there in full; this section adds the LOSO jackknife detail not shown in the main text.

Channel	2nd-order p range (24 reruns)	Most influential (2nd-order)	Pooled-perm p range	N reruns pooled-perm $p < 0.05$
`phi_resp_peak`	[0.0000, 0.0000]	`sub-xn017` (weakest $p =$ 0.0000)	[0.0020, 0.0020]	24/24
`phi_cardiac_peak`	[0.0014, 0.0105]	`sub-xn013` (weakest $p =$ 0.0105)	[0.0040, 0.0679]	22/24

Table 4: Leave-one-subject-out jackknife, 24 reruns per channel (one per dropped subject); pooled-permutation reruns use $n_{\text{perm}} = 500$ (reduced from 2000 for the 24×-repeated loop, S1.4). Respiratory peak-phase locking survives every single-subject removal on both tests; cardiac peak-phase locking loses pooled significance in 2/24 reruns. Source: `results/rvc_sleep_20260710/12_richsleep_maalb_peak_phase_deepdive/check3_leave_one_subject_out.csv`.

S2.4 Complete peak-phase alignment-moderation family (18 tests, main text R6)

Delta mode	Endp.	N	coef	p	FDR q	slope t-test p	slope Wilcoxon p	slope perm p
event_control	m	172	−0.029	0.578	0.722	0.252	0.330	0.271
event_control	d	172	+0.137	0.061	0.182	0.260	0.312	0.250
event_control	e	172	+0.078	0.333	0.500	0.662	0.123	0.673
event_control	d_n	172	+0.164	0.075	0.192	0.457	0.202	0.465
event_control	d_l	172	+0.125	0.090	0.203	0.121	0.123	0.099
event_control	d_s	172	+0.035	0.734	0.777	0.391	0.546	0.429
event_control	e_e	172	+0.036	0.704	0.777	0.563	0.349	0.573
event_control	e_s	172	+0.107	0.258	0.423	0.658	0.784	0.728
event_control	e_m	172	+0.035	0.602	0.722	0.086	0.105	0.084
event_prefar	m	160	−0.015	0.854	0.854	0.585	0.522	0.794
event_prefar	d	160	+0.243	0.0023	∗ 0.021 ∗	0.077	0.044	0.068
event_prefar	e	160	+0.136	0.035	0.139	0.313	0.674	0.369
event_prefar	d_n	160	+0.353	0.036	0.139	0.200	0.114	0.186
event_prefar	d_l	160	+0.184	0.0003	∗ 0.005 ∗	0.105	0.014	0.100
event_prefar	d_s	160	+0.218	0.182	0.327	0.445	0.388	0.462
event_prefar	e_e	160	+0.028	0.552	0.722	0.573	0.648	0.553
event_prefar	e_s	160	+0.286	0.039	0.139	0.355	0.756	0.410
event_prefar	e_m	160	+0.133	0.111	0.222	0.143	0.097	0.148

Table 5: Complete 18-test resp_peak_alignment moderation family (WLS-cluster, weights $1/n_{\text{events,subject}}$, cluster-robust SE by subject; 20/24 subjects contribute a per-subject slope, `min_events=3`). Bold ***italic*** FDR q = BH-FDR survivor. `event_control` mode is null throughout (all $q \geq 0.18$); `event_prefar` mode has two survivors (`d`, `d_l`), both with the slope Wilcoxon test agreeing but the slope t-test and sign-flip permutation test not fully agreeing (see main text R6, `S3`). Source: `results/rvc_sleep_20260710/13_peak_phase_alignment_moderation/resp_peak_alignment_moderation.csv`.

S2.5 Secondary circular-regression sensitivity check (18 rows, exploratory, not FDR-corrected)

Delta mode	Endp.	cos p	sin p	joint Wald p	geom.-optimal phase	dist. to $\mu_{\text{resp,peak}}$
event_control	m	0.520	0.779	0.791	$+34.1°$	$162.5°$
event_control	d	0.693	0.018	0.058	$-97.3°$	$31.1°$
event_control	e	0.959	0.098	0.208	$-92.0°$	$36.3°$
event_control	d_n	0.999	0.076	0.174	$-90.0°$	$38.3°$
event_control	d_l	0.372	0.032	0.055	$-109.0°$	$19.4°$
event_control	d_s	0.766	0.423	0.620	$-70.4°$	$58.0°$
event_control	e_e	0.915	0.544	0.797	$-101.9°$	$26.5°$
event_control	e_s	0.868	0.120	0.292	$-85.3°$	$43.1°$
event_control	e_m	0.803	0.355	0.591	$-77.8°$	$50.6°$
event_prefar	m	0.252	0.377	0.488	$+129.8°$	$101.9°$
event_prefar	d	0.473	0.0044	0.0113	$-97.2°$	$31.1°$
event_prefar	e	0.429	0.049	0.108	$-101.0°$	$27.4°$
event_prefar	d_n	0.429	0.021	0.070	$-81.3°$	$47.1°$
event_prefar	d_l	0.122	0.0013	0.0008	$-117.4°$	$10.9°$
event_prefar	d_s	0.248	0.364	0.340	$-123.0°$	$5.4°$
event_prefar	e_e	0.278	0.949	0.491	$+176.3°$	$55.3°$
event_prefar	e_s	0.854	0.030	0.094	$-92.3°$	$36.1°$
event_prefar	e_m	0.402	0.214	0.272	$-112.2°$	$16.1°$

Table 6: delta $\sim \cos(\varphi_{\text{resp,peak}}) + \sin(\varphi_{\text{resp,peak}}) +$ covariates, WLS-cluster, exploratory and not part of the primary 18-test family (`S1.5`). "Geom.-optimal phase" $= \operatorname{atan2}(\text{sin_coef}, \text{cos_coef})$, the phase direction that maximizes the fitted delta; "dist. to $\mu_{\text{resp,peak}}$" is its wrapped angular distance from the independently-estimated held-out population-preferred phase ($-128.4°$, S2.6). `event_prefar d_l` (main text's strongest alignment-moderation row) has the smallest such distance ($10.9°$) of any of the 18 rows. Source: `results/rvc_sleep_20260710/13_peak_phase_alignment_moderation/resp_peak_circular_regression_secondary.csv`.

S2.6 Held-out population-preferred phase estimate

$\mu_{\text{resp,peak}}$	R (2nd-order)	N subj.	N held-out events	N excluded (geom.-valid)
$-128.4°$ (-2.2404 rad)	0.6657	24	22,839	190

Table 7: Held-out, subject-balanced second-order circular mean of `phi_resp_peak`, computed on the full catalog **excluding** the 190 geometry-valid events used in R6′s regression (S1.5), avoiding circularity. Consistent with the full-catalog (non-held-out) pooled mean angle reported in main text R4 ($-122.6°$) - the $< 1\%$ exclusion changes the estimate negligibly. Source: `results/rvc_sleep_20260710/13_peak_phase_alignment_moderation/held_out_mu_resp_peak.csv`.

Bibliography